IsdaSafe Admin

Admin Panel of the Pond Water Quality Monitoring System

System analysis & design diagrams - admin side (companion to the farmer app's IsdaSafe System Diagrams)

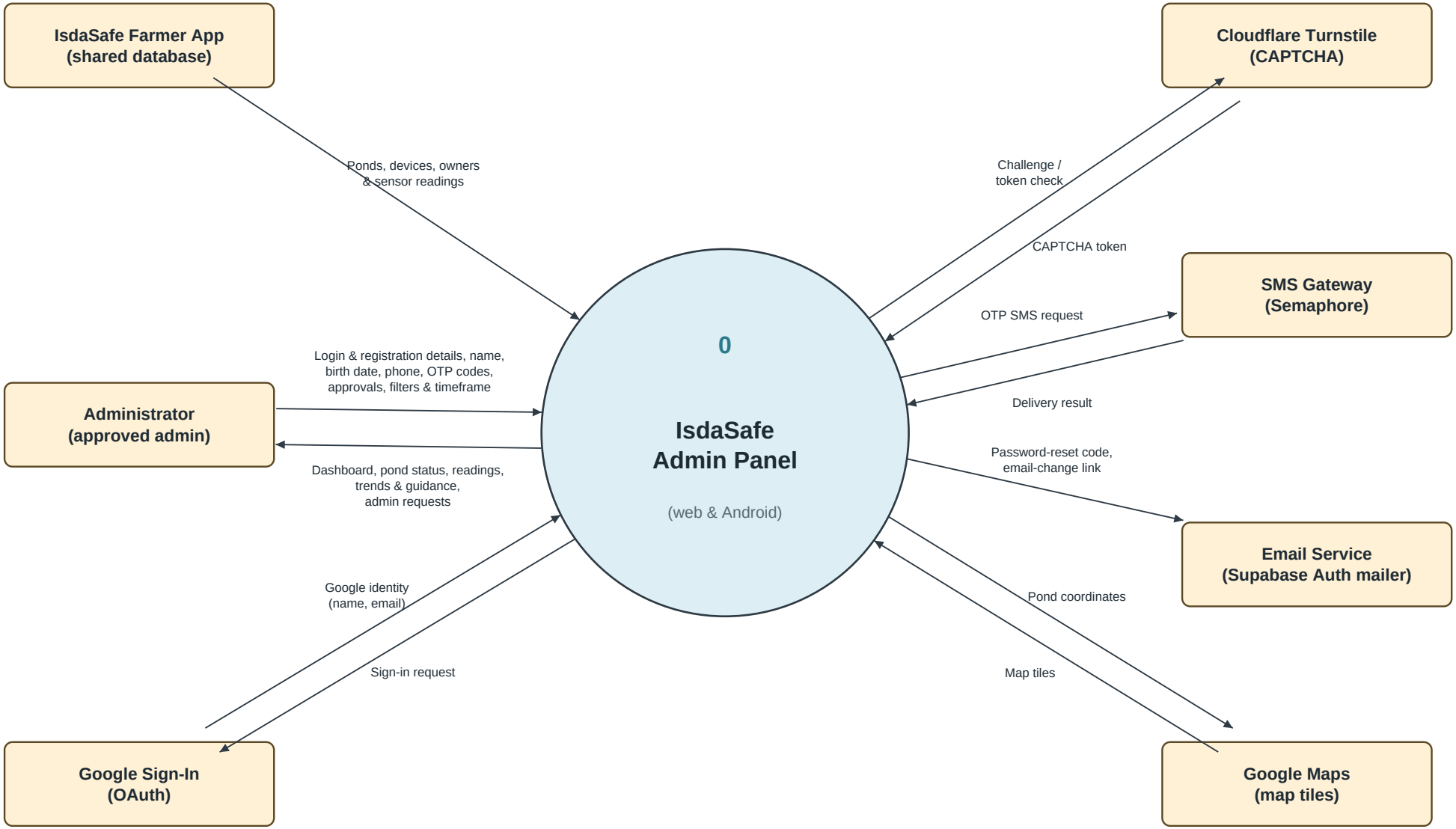
1. **Context Diagram**
   The admin panel as one process and the outside parties it talks to.
2. **Level 0 Data Flow Diagram**
   The admin panel broken into its five major processes and data stores.
3. **Child Diagrams (Processes 1.0 - 5.0)**
   Each Level 0 process exploded into its sub-processes.
4. **State Transition Diagrams**
   Life cycle of an admin account and of an admin's session at the login gate.
5. **Flowchart**
   The whole panel step by step: register, verify, approval, monitoring, trends, settings.
6. **Entity Relationship Diagram**
   The admin tables and the shared farmer tables the panel reads, and how they relate.
7. **Data Dictionary**
   Every column of the admin tables, plus the database functions and triggers.
8. **Software Block Diagram**
   The admin software split into its major blocks: app, cloud backend and outside services.
9. **Software Schematic Diagram**
   Every screen, service, backend function and outside service wired together, plus an interface list.

Context diagram: Yourdon/DeMarco notation. Level 0 and child DFDs: Gane & Sarson notation. ERD: crow's-foot notation. Block & schematic diagrams: software architecture view.

The admin panel shares the IsdaSafe Supabase project with the farmer app: farmer tables are written by the farmer app and only read here.

1. Context Diagram

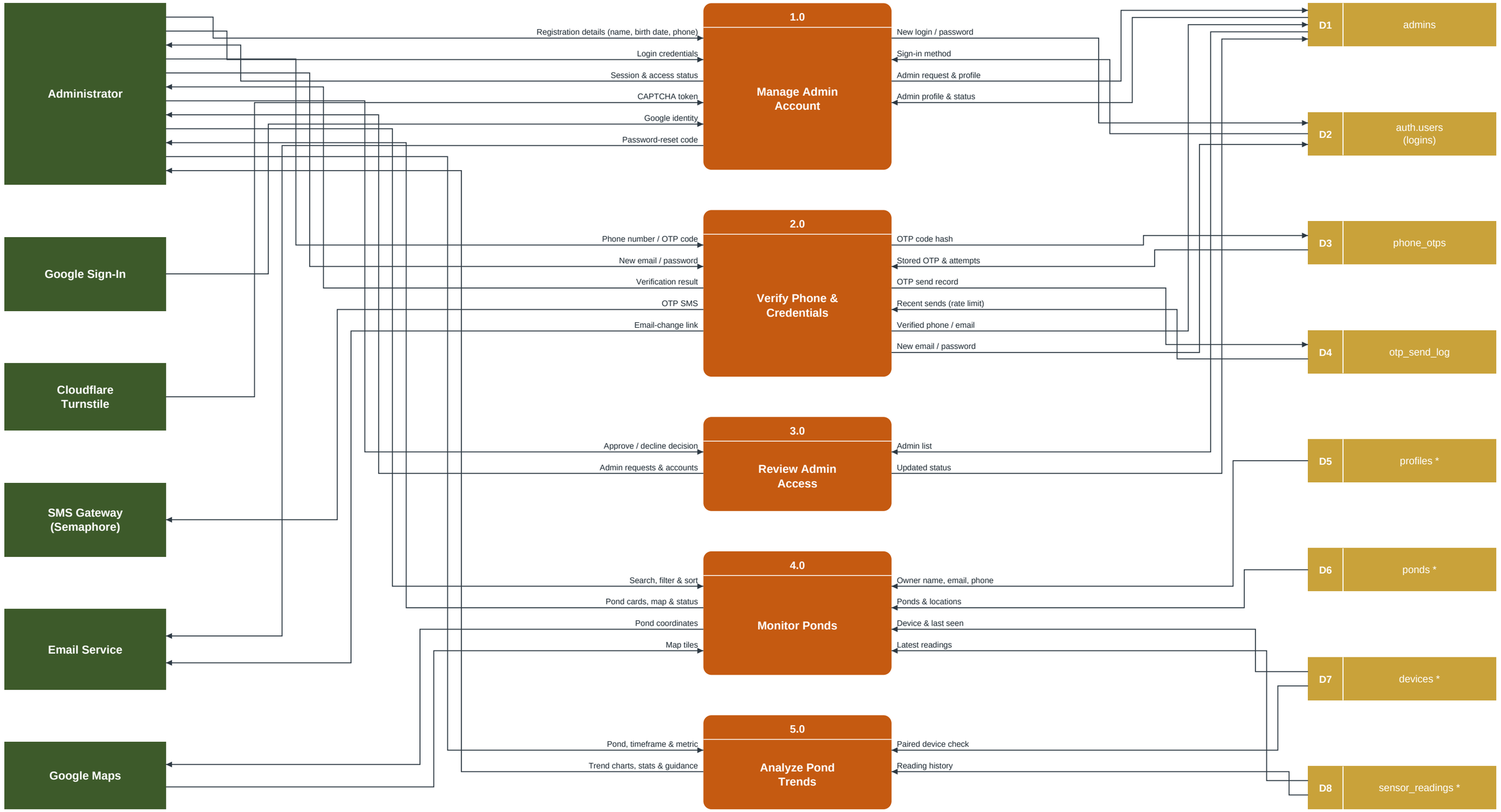
Shows the IsdaSafe Admin Panel as a single process (0) and every external entity that sends data to it or receives data from it.



External entity Process

2. Level 0 Data Flow Diagram

*Process 0 exploded into its five major processes (Gane & Sarson notation). Green = external entity, orange = process, gold = data store. * = farmer-app table, read only by the admin panel.*

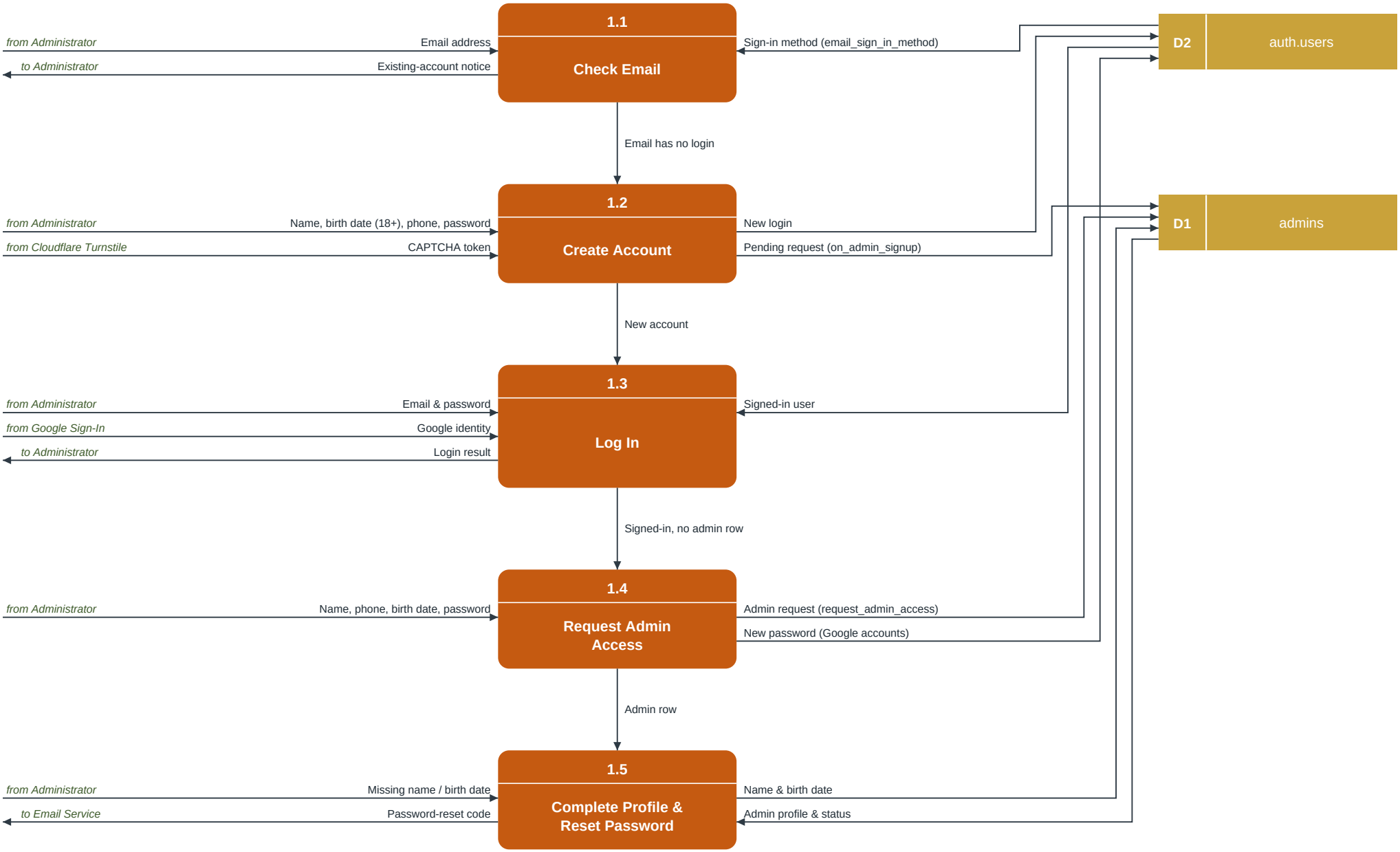


** D5-D8 are written by the IsdaSafe farmer app (see its own diagrams). The admin panel reads them only through admin_* database functions that first check the caller is an approved admin.*

External entity Process Data store

3. Child Diagram - Process 1.0: Manage Admin Account

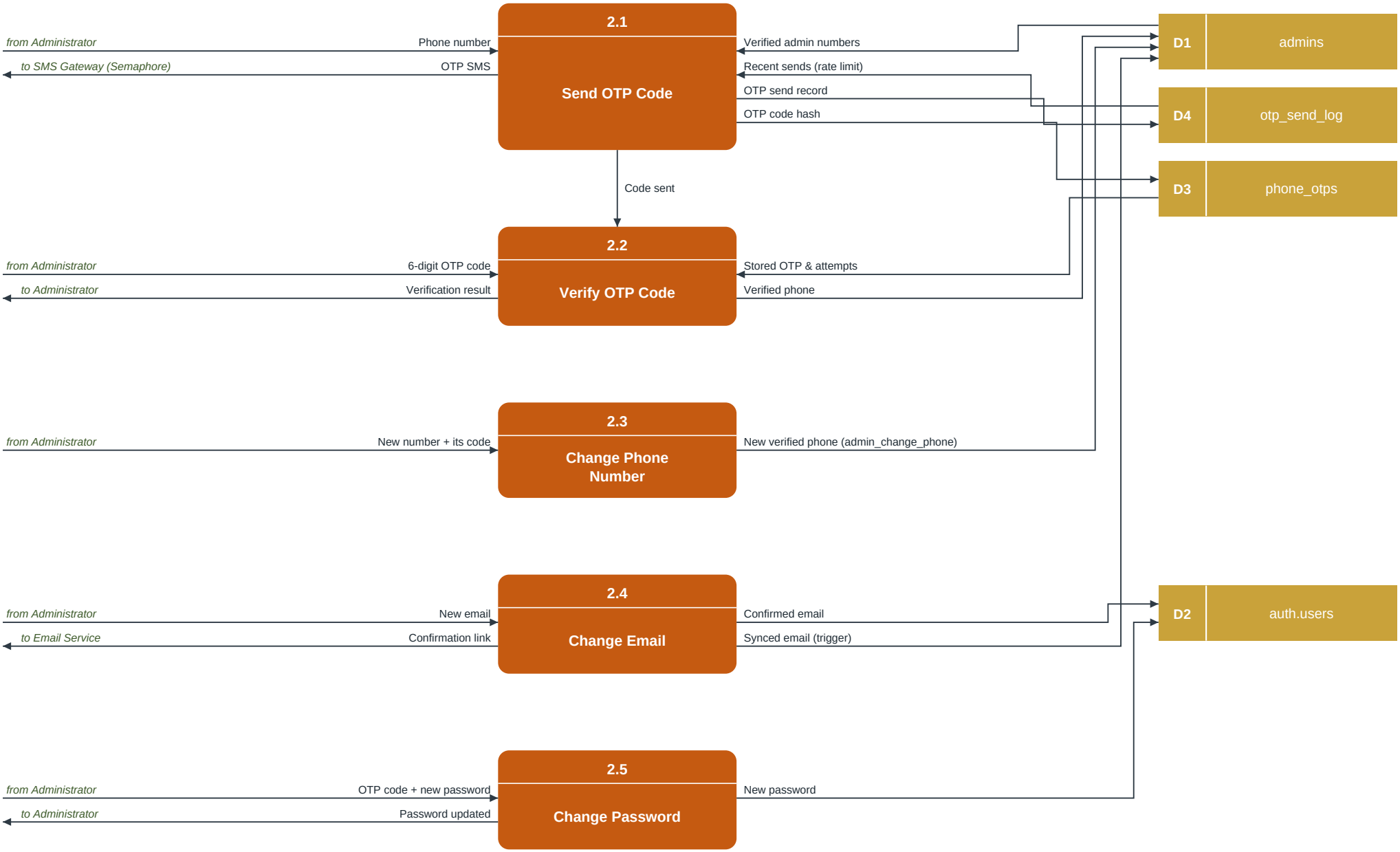
Explodes process 1.0 of the Level 0 DFD. External entities are not drawn: each arrow from the left edge names its entity at the open end.



Process Data store

3. Child Diagram - Process 2.0: Verify Phone & Credentials

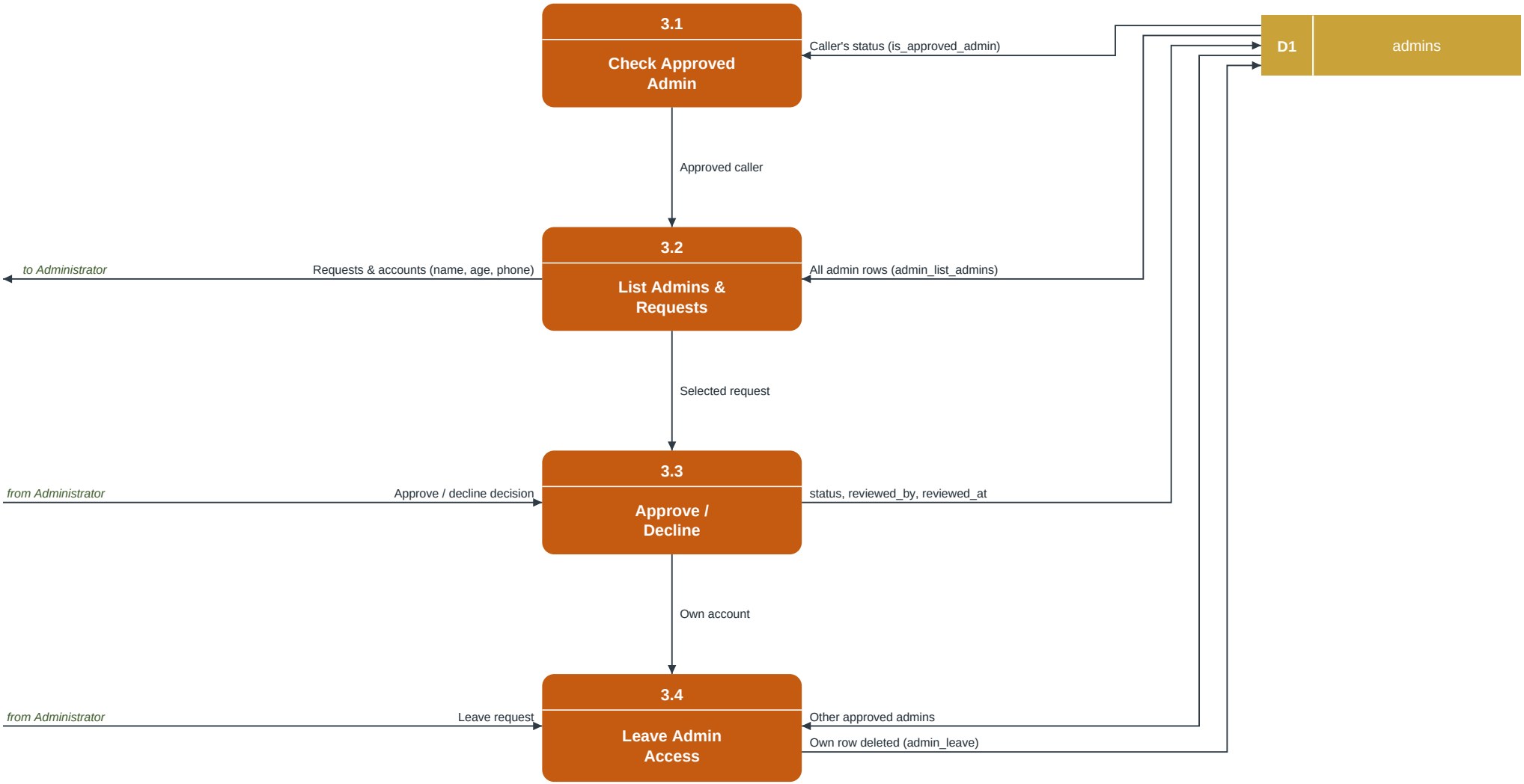
Explodes process 2.0. Codes are texted by the shared send-semaphore-otp Edge Function and checked in the database, never on the device.



Process Data store

3. Child Diagram - Process 3.0: Review Admin Access

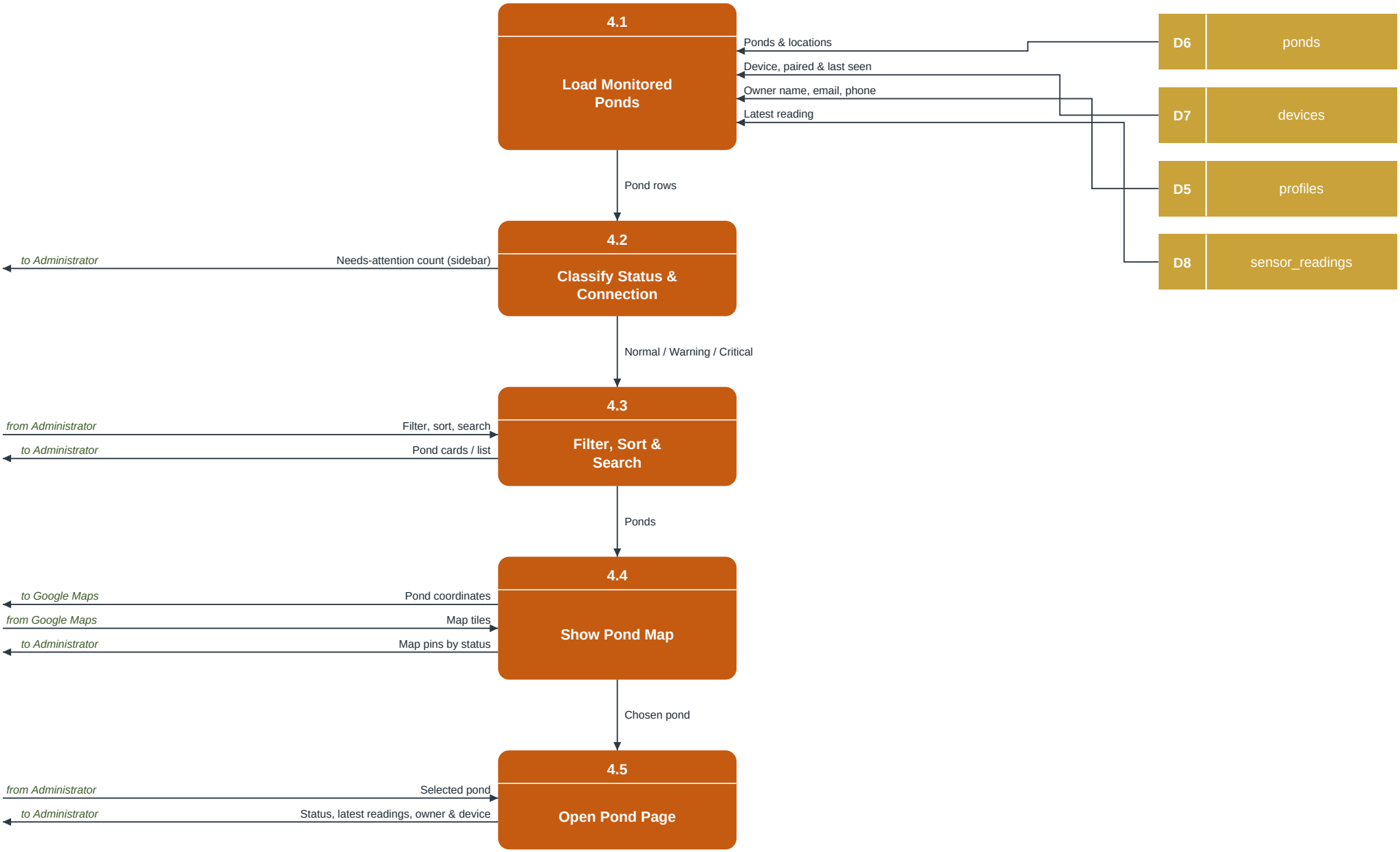
Explodes process 3.0. Every function first checks is_approved_admin(); nobody can change their own access, and the last approved admin can't leave.



Process Data store

3. Child Diagram - Process 4.0: Monitor Ponds

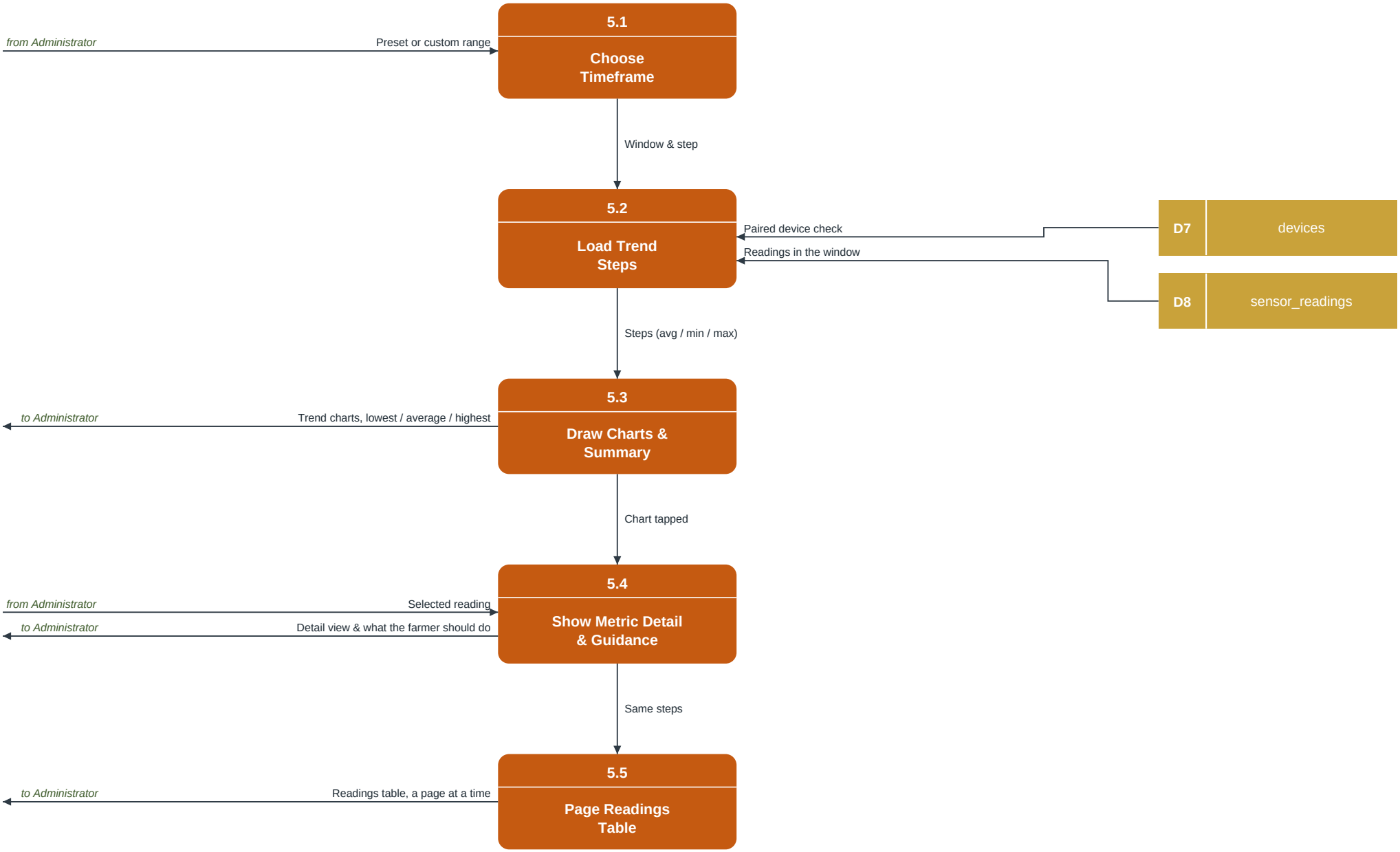
Explodes process 4.0. admin_monitored_ponds() joins every pond that has a paired device with its owner and latest reading.



Process Data store

3. Child Diagram - Process 5.0: Analyze Pond Trends

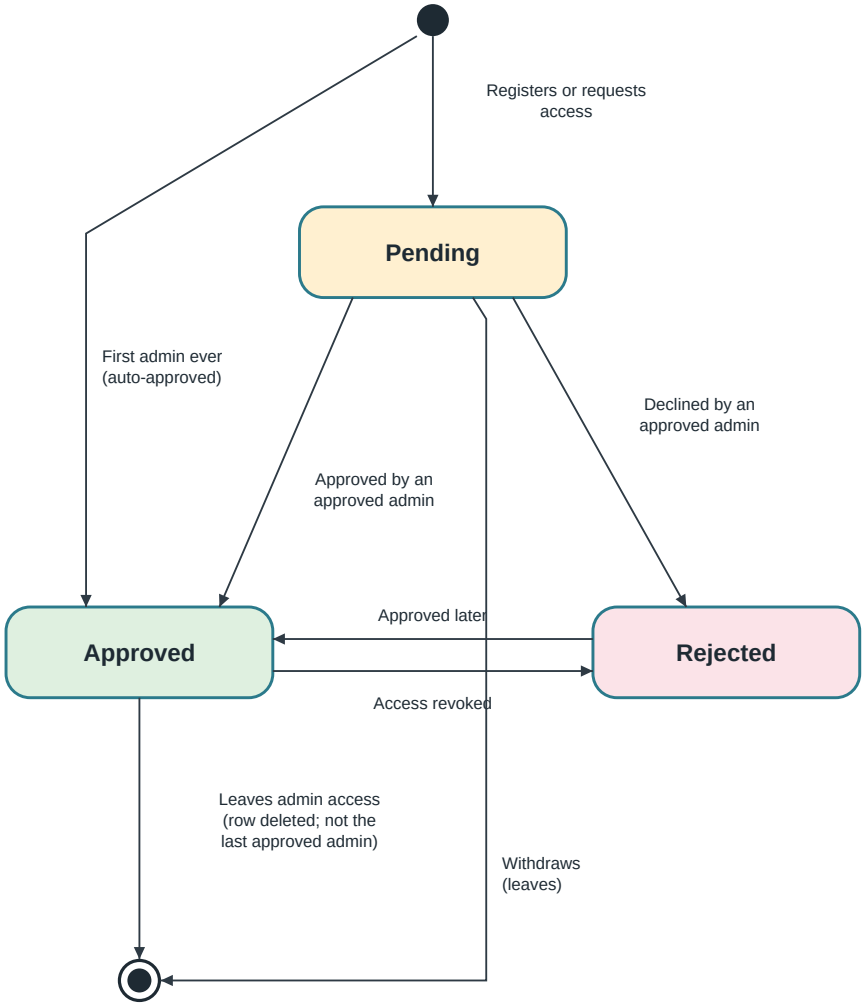
Explodes process 5.0. admin_pond_reading_trend() groups readings into 15-minute / hour / day / week / month / year steps on the server.



4. State Transition Diagrams

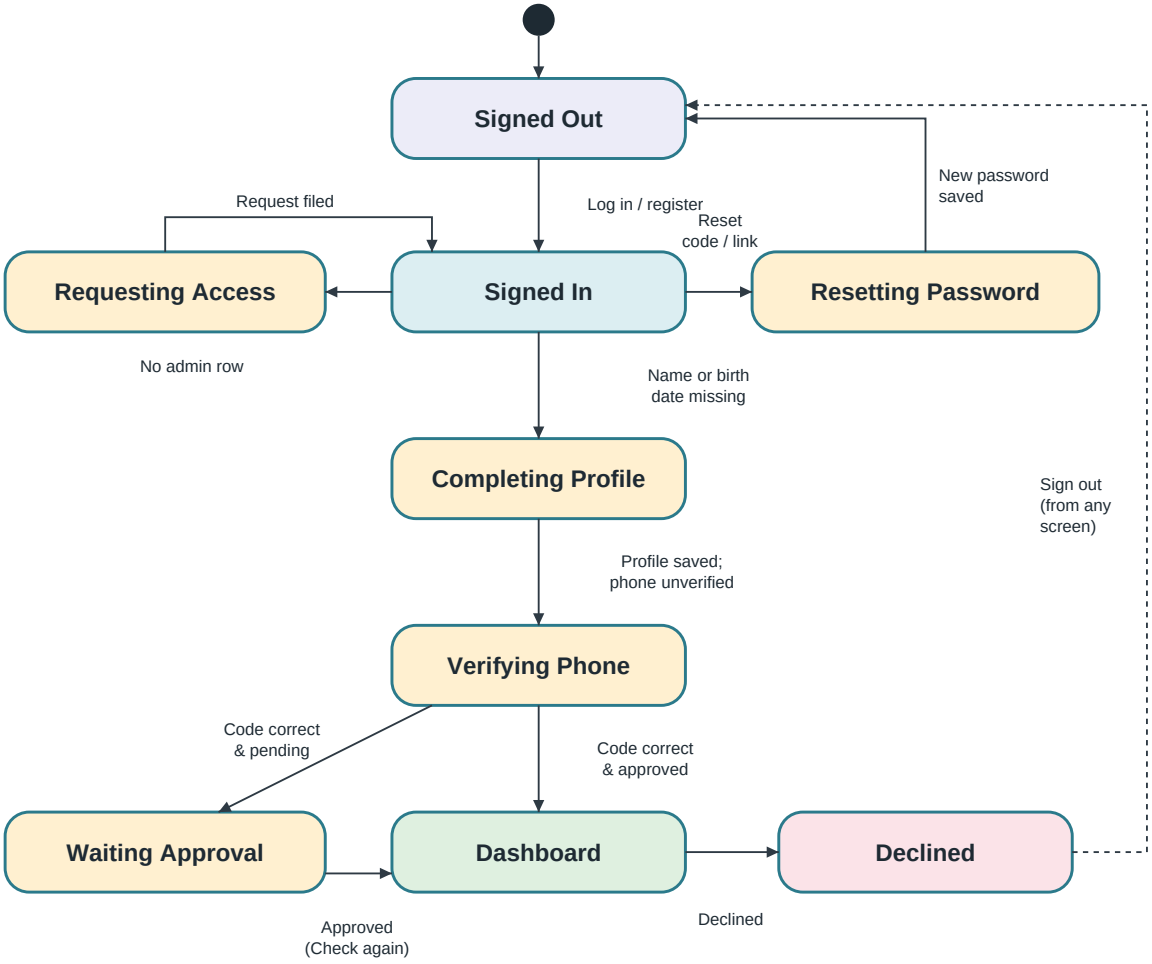
Left: life cycle of an admin account (its admins.status). Right: the screens AdminGate walks a signed-in admin through.

A. Admin Account



A declined account can't clear its row and request again; another admin must approve it.

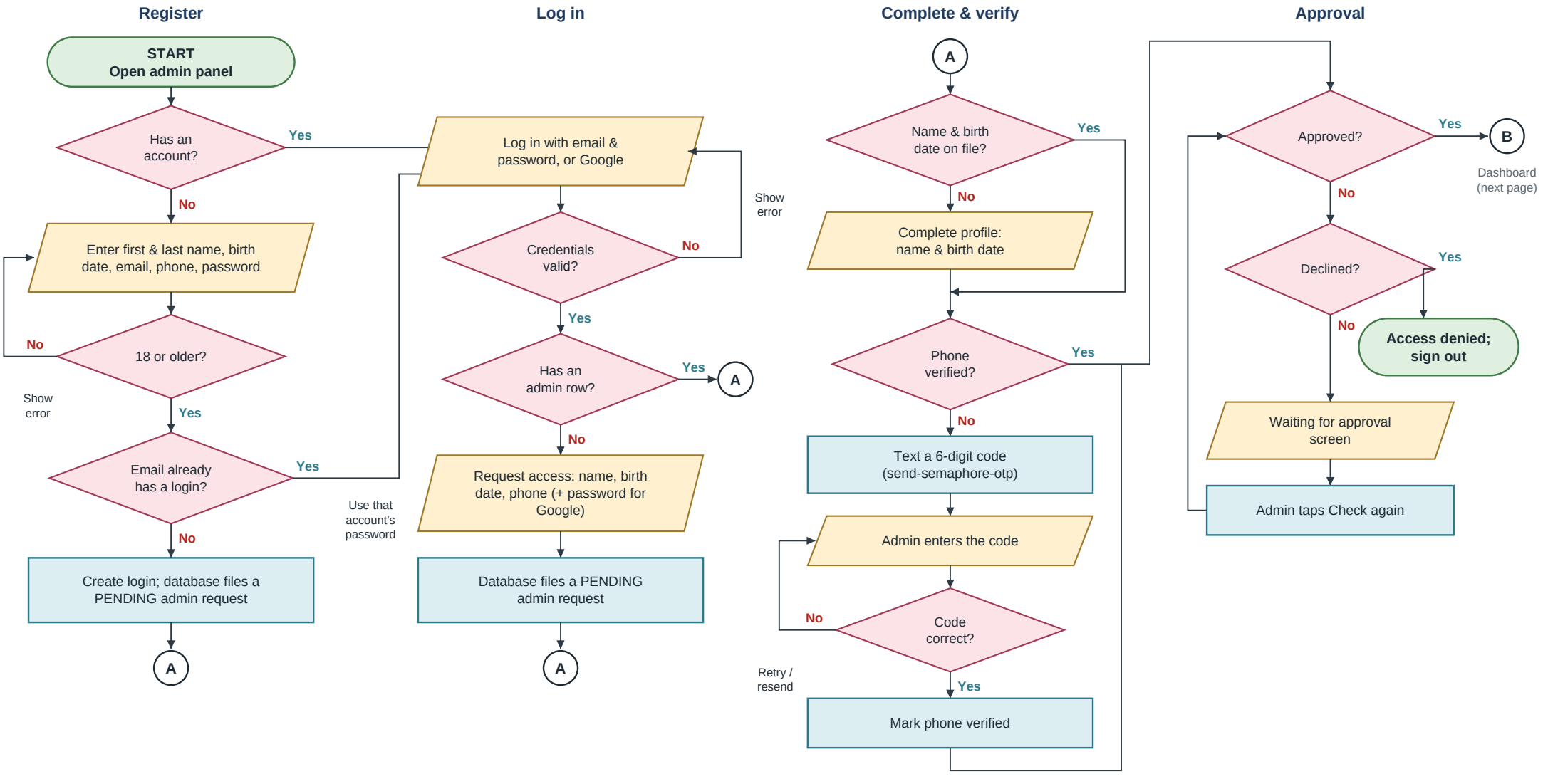
B. Admin Session (login gate)



Event / action notation: "trigger / what the system does".
Each screen is chosen again on every sign-in from the admin row (AdminGate).

5. Flowchart (1 of 2) - Register, Log In & Get Approved

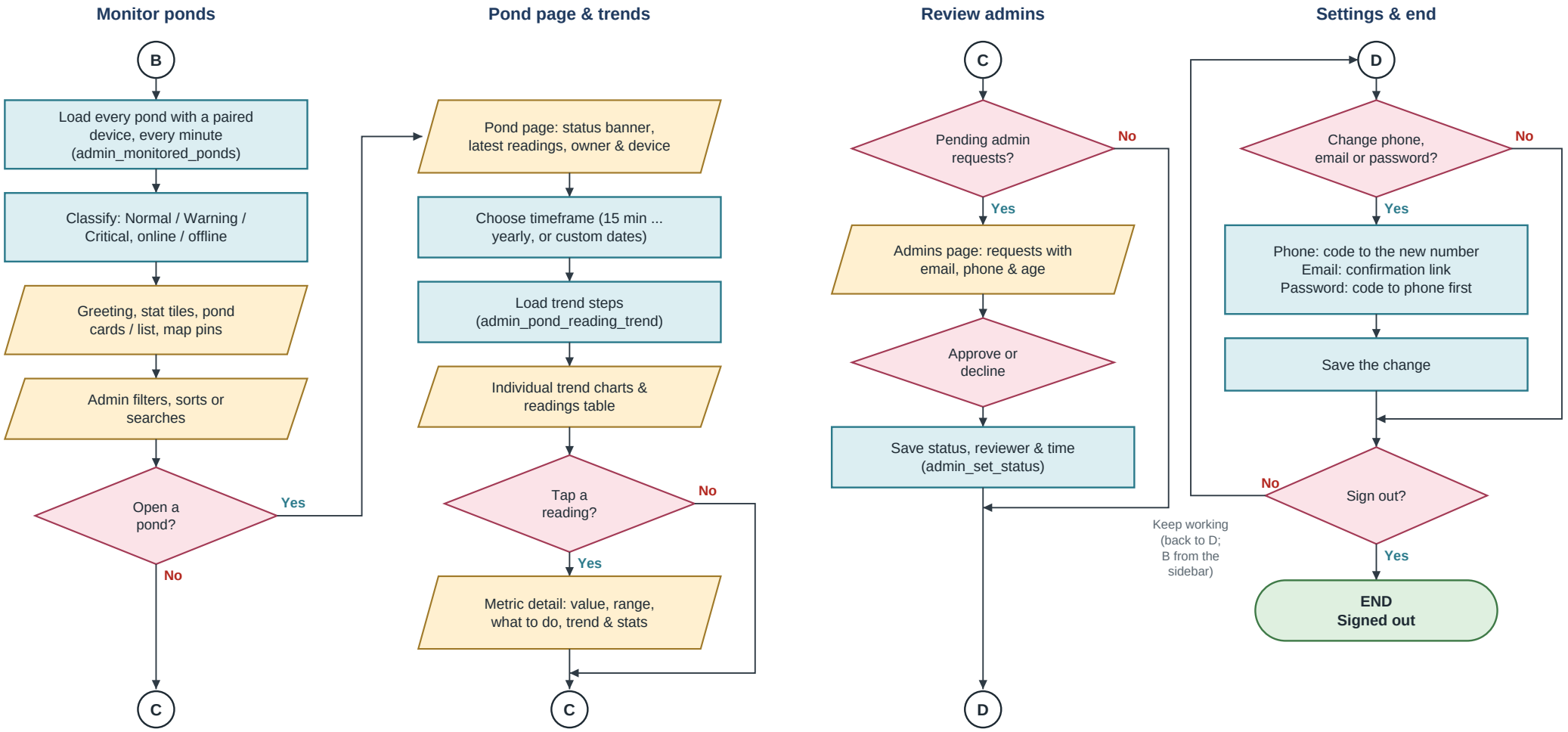
From opening the admin panel to reaching the dashboard. Read top to bottom, left to right.



Start / End · Process · Input / Output · Decision · Connector (same letter = same point)

5. Flowchart (2 of 2) - Monitoring, Trends, Admins & Settings

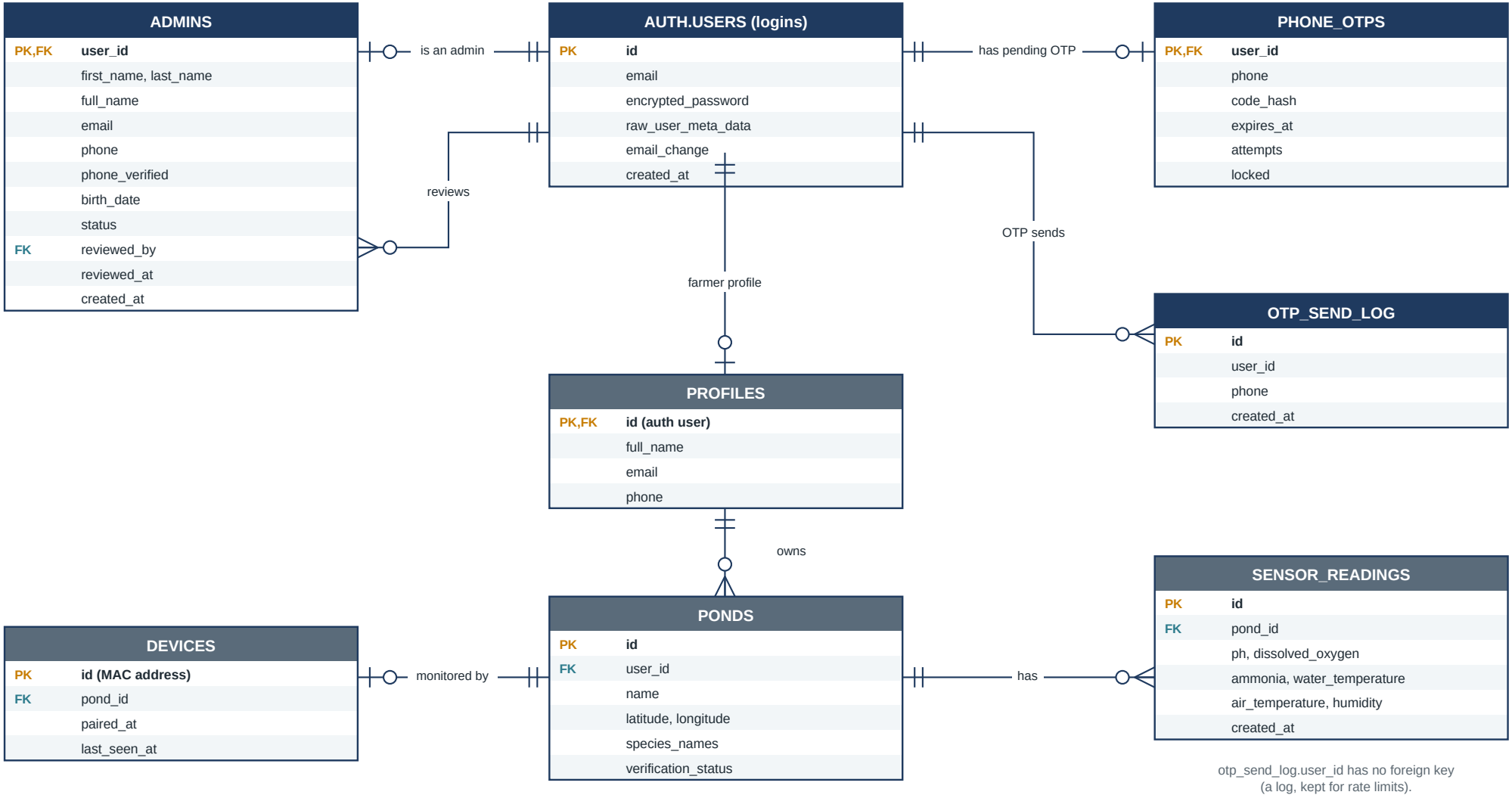
What an approved admin does in the panel, until signing out. Read top to bottom, left to right.



Start / End | Process | Input / Output | Decision | Connector (same letter = same point)

6. Entity Relationship Diagram

Admin tables (navy) and the shared farmer tables the panel reads (grey). Crow's-foot notation: | = one, < = many, o = optional. Key columns only - see the Data Dictionary.



otp_send_log.user_id has no foreign key (a log, kept for rate limits).

The panel only reads the grey tables, through admin_monitored_ponds() and admin_pond_reading_trend().

Admin table Farmer table (read only)

7. Data Dictionary

Admin tables in the public schema of the shared IsdaSafe Supabase database. PK = primary key, FK = foreign key, timestamptz = timestamp with time zone.

Table 7.1 admins

One row per admin account or admin request (created at sign-up or by request_admin_access).

Column	Data type	Null	Key	Default	Description
user_id	uuid	No	PK, FK	-	Login; references auth.users.id (deleted with the login)
first_name	text	Yes	-	-	First name (null for admins from before it was asked)
last_name	text	Yes	-	-	Last name
full_name	text	No	-	-	"first last", kept in step by the functions; shown in lists
email	text	No	-	-	Login email; updated by a trigger when an email change is confirmed
phone	text	Yes	-	-	PH mobile in E.164 (+639XXXXXXXXX) for OTP codes; may match a farmer's
phone_verified	boolean	No	-	false	True once an OTP sent to phone was confirmed (required to use the panel)
birth_date	date	Yes	-	-	Date of birth; must be 18 or older (is_adult)
status	text	No	-	'pending'	'pending', 'approved' or 'rejected'; the first admin ever is approved
reviewed_by	uuid	Yes	FK	-	Admin who last approved / declined it; references auth.users.id
reviewed_at	timestamptz	Yes	-	-	When it was last reviewed
created_at	timestamptz	No	-	now()	When the request was filed

Table 7.2 phone_otps

The one OTP code currently waiting per login (shared with the farmer app).

Column	Data type	Null	Key	Default	Description
user_id	uuid	No	PK, FK	-	Login the code belongs to; references auth.users.id
phone	text	No	-	-	Number the code was texted to
code_hash	text	No	-	-	SHA-256 hash of the code (never stored plain)
expires_at	timestamptz	No	-	-	5 minutes after sending
attempts	integer	No	-	0	Wrong guesses so far
locked	boolean	No	-	false	True after 5 wrong guesses

7. Data Dictionary (continued)

Admin-side tables and the shared tables the panel reads.

Table 7.3 otp_send_log

Every OTP SMS sent; limits sends to 1/min, 3 per 10 min and 8 per day per login and number.

Column	Data type	Null	Key	Default	Description
id	bigint	No	PK	identity	Auto-increment log ID
user_id	uuid	No	-	-	Login that requested the code
phone	text	No	-	-	Number the code was sent to
created_at	timestamptz	No	-	now()	When it was sent

Table 7.4 auth.users (Supabase Auth)

Shared logins of admins and farmers; managed by Supabase Auth. Columns the admin side relies on.

Column	Data type	Null	Key	Default	Description
id	uuid	No	PK	-	Login ID (admins.user_id)
email	text	Yes	-	-	Login email (copied to admins.email)
encrypted_password	text	Yes	-	-	Empty for Google-only logins (email_sign_in_method returns 'oauth')
raw_user_meta_data	jsonb	Yes	-	-	Sign-up data: admin_signup, first_name, last_name, birth_date, admin_phone
email_change	text	Yes	-	-	New email waiting for its confirmation link

Table 7.5 Farmer tables read by the panel

Written by the IsdaSafe farmer app; see its Data Dictionary for every column.

Table	Columns read	Read through
profiles	full_name, email, phone	admin_monitored_ponds() (pond owner)
ponds	id, name, latitude, longitude, species_names, verification_status, user_id	admin_monitored_ponds()
devices	id, pond_id, paired_at, last_seen_at	admin_monitored_ponds(), admin_pond_reading_trend()
sensor_readings	ph, dissolved_oxygen, ammonia, water/air temperature, humidity, created_at	latest per pond; trend steps (avg / min / max)

7. Data Dictionary (continued) - Functions & Triggers

SECURITY DEFINER functions the panel calls (PostgREST RPC). Admin-only ones check is_approved_admin() first.

Table 7.6 Database functions

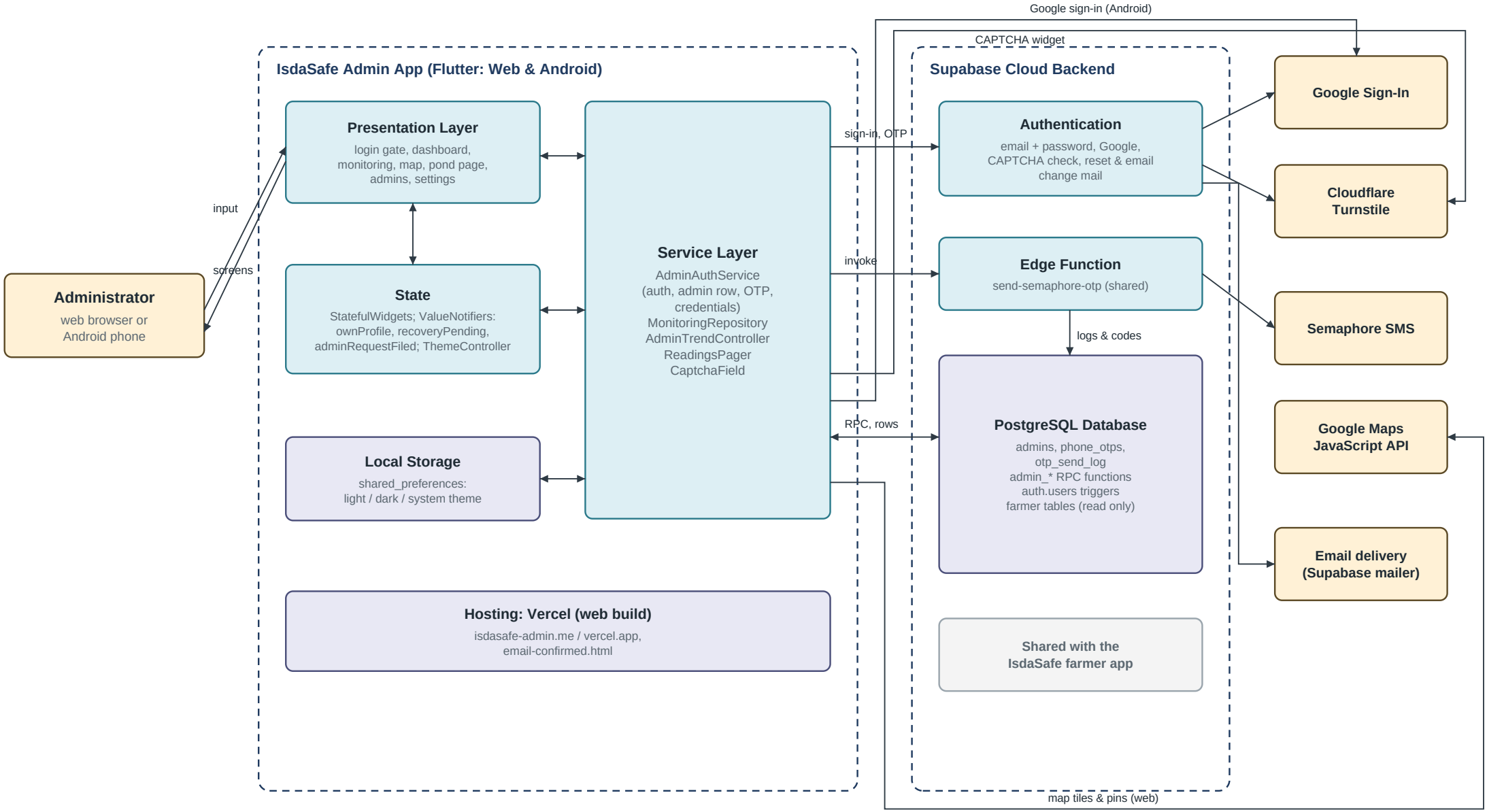
Function	Arguments	Returns	Who may call	Purpose
is_approved_admin	-	boolean	signed in	Whether the caller has an approved admins row
request_admin_access	first, last, phone, birth_date	text	signed in	Files a request (or fills a missing name / birth date)
admin_set_own_name	first, last	void	signed in	Sets the caller's first / last / full name
admin_set_own_birth_date	birth_date	void	signed in	Sets the caller's birth date (18+, not in the future)
admin_set_own_phone	phone	void	signed in	Sets an unverified number before the first code (login gate)
admin_verify_phone_otp	code	text	signed in	Checks the code for the number on file; null = success
admin_check_phone_available	phone	void	signed in	Refuses a number another verified admin has
admin_change_phone	code	text	signed in	Swaps in a new number once its code is right
admin_list_admins	-	setof admins	approved	Every admin and request, newest first
admin_set_status	user_id, status	void	approved	Approves / declines another admin (never self)
admin_leave	-	void	signed in	Deletes own row (not the last approved admin)
admin_monitored_ponds	-	table	approved	Ponds with a paired device, owner & latest reading
admin_pond_reading_trend	pond_id, start, end, step	table	approved	Readings grouped per step: avg / min / max / count
email_has_account	email	boolean	anyone	Whether a login uses the email (Register)
email_sign_in_method	email	text	anyone	null / 'password' / 'oauth' (Google-only)
is_adult	birth_date	boolean	anyone	18 or older today (Manila time)

Table 7.7 Triggers

Trigger	On	Function	What it does
on_admin_signup	auth.users INSERT	handle_admin_signup	Sign-ups with admin_signup metadata become a pending admins row (18+)
on_auth_email_change_sync_admin	auth.users UPDATE OF email	sync_admin_email	Copies a confirmed new email onto admins.email

8. Software Block Diagram

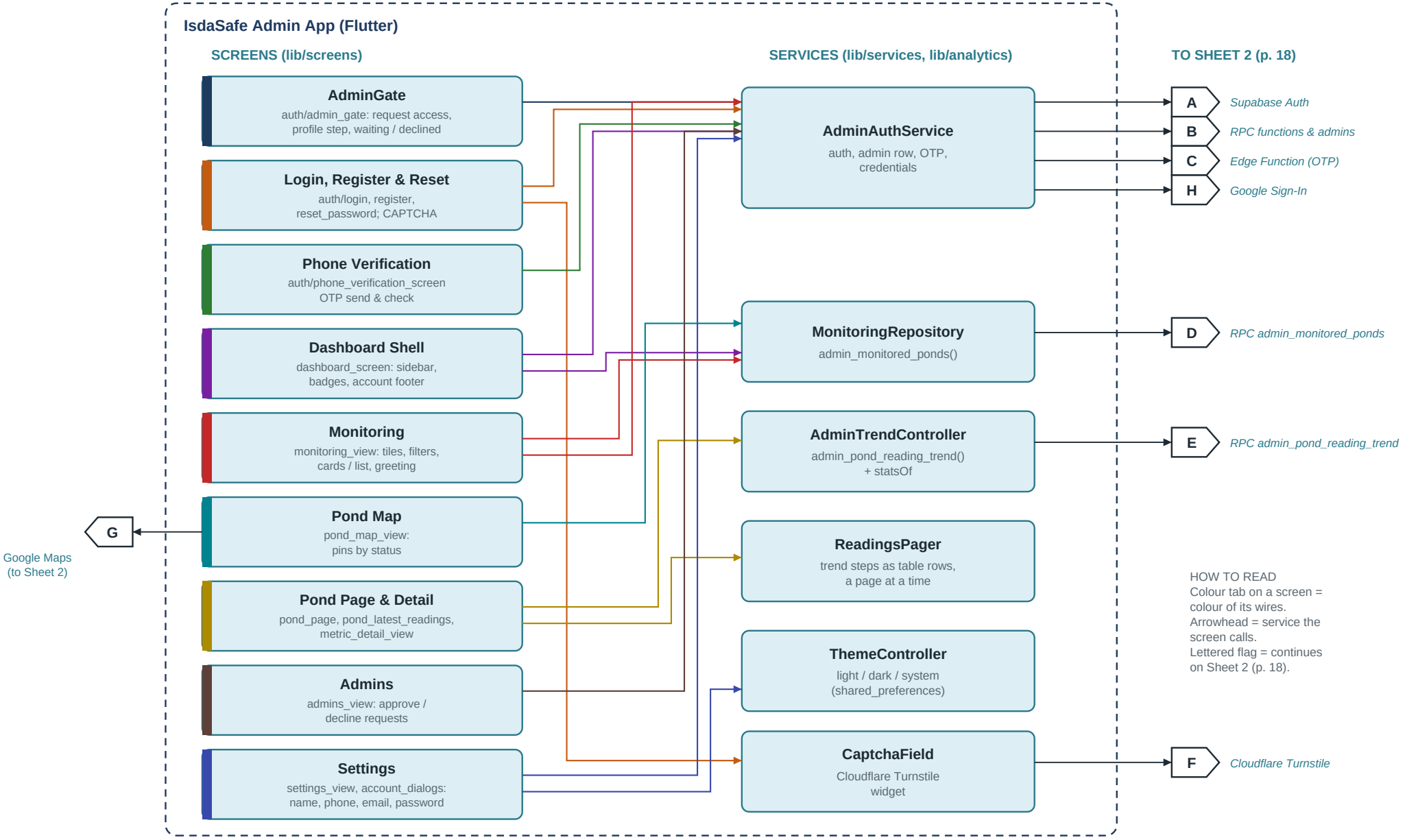
The admin software as its major blocks and the main data passed between them. Dashed outlines group the blocks that run on the same platform.



User / outside service · Software block · Data store / hosting · Platform (runs together)

9. Software Schematic (Sheet 1 of 2) - Screens & Services

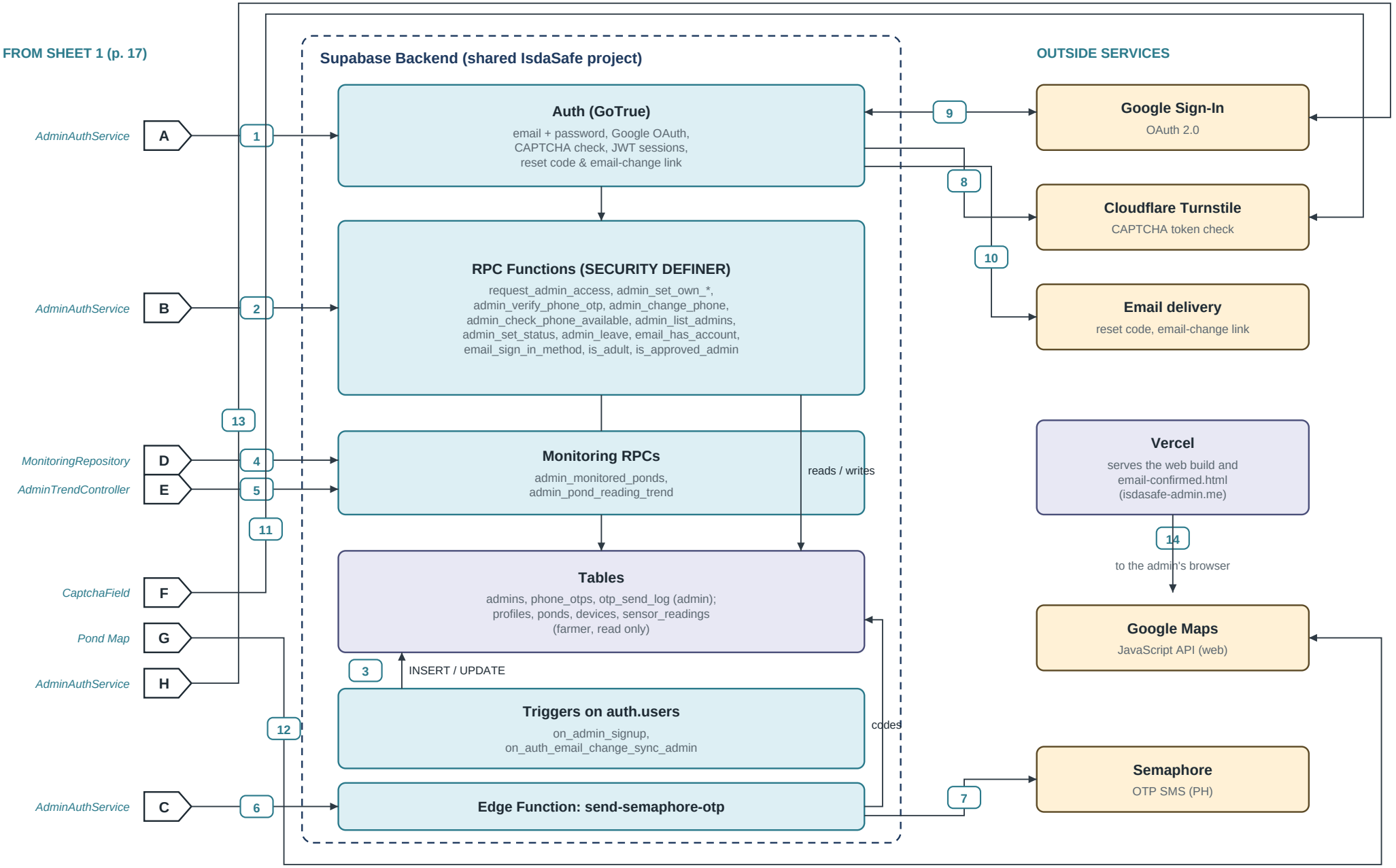
Each screen has its own colour: follow it to the arrowheads to see which services it uses. Lettered flags continue on Sheet 2.



Software module Platform Screen wire (colour per screen)

9. Software Schematic (Sheet 2 of 2) - Backend & Cloud

Every wire entering from the left comes from Sheet 1 (p. 17) through the flag with the same letter. Numbers match the Interface List on the next page.



Outside service | Software module | Data store / hosting | Platform

9. Software Schematic Diagram (continued) - Interface List

Each numbered wire of the schematic (Sheets 1-2): the two modules it joins, the interface or protocol used, and the data carried.

#	From	To	Interface / protocol	Data carried
1	AdminAuthService	Supabase Auth	supabase_flutter (HTTPS, JWT)	sign-up / sign-in, Google, sessions, CAPTCHA token, reset, email change
2	AdminAuthService	RPC functions, admins	PostgREST RPC / select (HTTPS)	admin requests, profile, OTP checks, approvals, own admin row
3	Triggers on auth.users	admins table	PL/pgSQL trigger	new admin request at sign-up; confirmed email copied over
4	MonitoringRepository	admin_monitored_ponds	PostgREST RPC (HTTPS)	ponds, owners, devices, latest readings
5	AdminTrendController	admin_pond_reading_trend	PostgREST RPC (HTTPS)	pond, window, step -> avg / min / max / count per metric
6	AdminAuthService	send-semaphore-otp	functions.invoke (HTTPS + JWT)	phone number -> code stored as a hash in phone_otps
7	send-semaphore-otp	Semaphore	HTTPS (api.semaphore.co)	6-digit OTP SMS to a PH mobile number
8	Supabase Auth	Cloudflare Turnstile	siteverify (HTTPS)	CAPTCHA token check on sign-up, log in and reset
9	Supabase Auth	Google Sign-In	OAuth 2.0	Google identity (name, email)
10	Supabase Auth	Email delivery	SMTP (Supabase mailer)	password-reset code, email-change confirmation link
11	CaptchaField	Cloudflare Turnstile	Turnstile widget (JS / WebView)	one-time CAPTCHA token
12	Pond Map	Google Maps	google_maps_flutter (Maps JS API)	pond pins; map tiles
13	AdminAuthService	Google Sign-In	OAuth redirect (web) / google_sign_in	Google account choice, ID token
14	Vercel	Admin's browser	HTTPS static hosting	web build, email-confirmed.html

Wires between screens and services are in-app Dart calls (State methods, ListenableBuilder / ValueListenableBuilder) and are not numbered.